MF

CASE STUDY

LEGAL / CRIMINAL DEFENSE

Casemap — AI System for Legal Case-File Analysis

An AI system for a criminal defense practice, combining multimodal document ingestion, classification and cited RAG to help lawyers analyze evidence across 3,000+ page case files.

KEY METRIC

28 hours*

less on document review, per large case

2x*
Large-case capacity per year

3,000+
Pages made searchable per case

* Estimated impact based on workflow analysis and pilot observations.

Manos Fragkiadakis

AI Engineer

LEGAL / CRIMINAL DEFENSE

28 hours less on document review.

PROJECT
Casemap

CLIENT
Criminal defense practice

INDUSTRY
Legal / Criminal Defense

ROLE
AI Engineer

CONTEXT
A 4-person criminal defense practice spending 60+ hours analyzing every large case file — thousands of pages of scanned, unstructured documents per case.

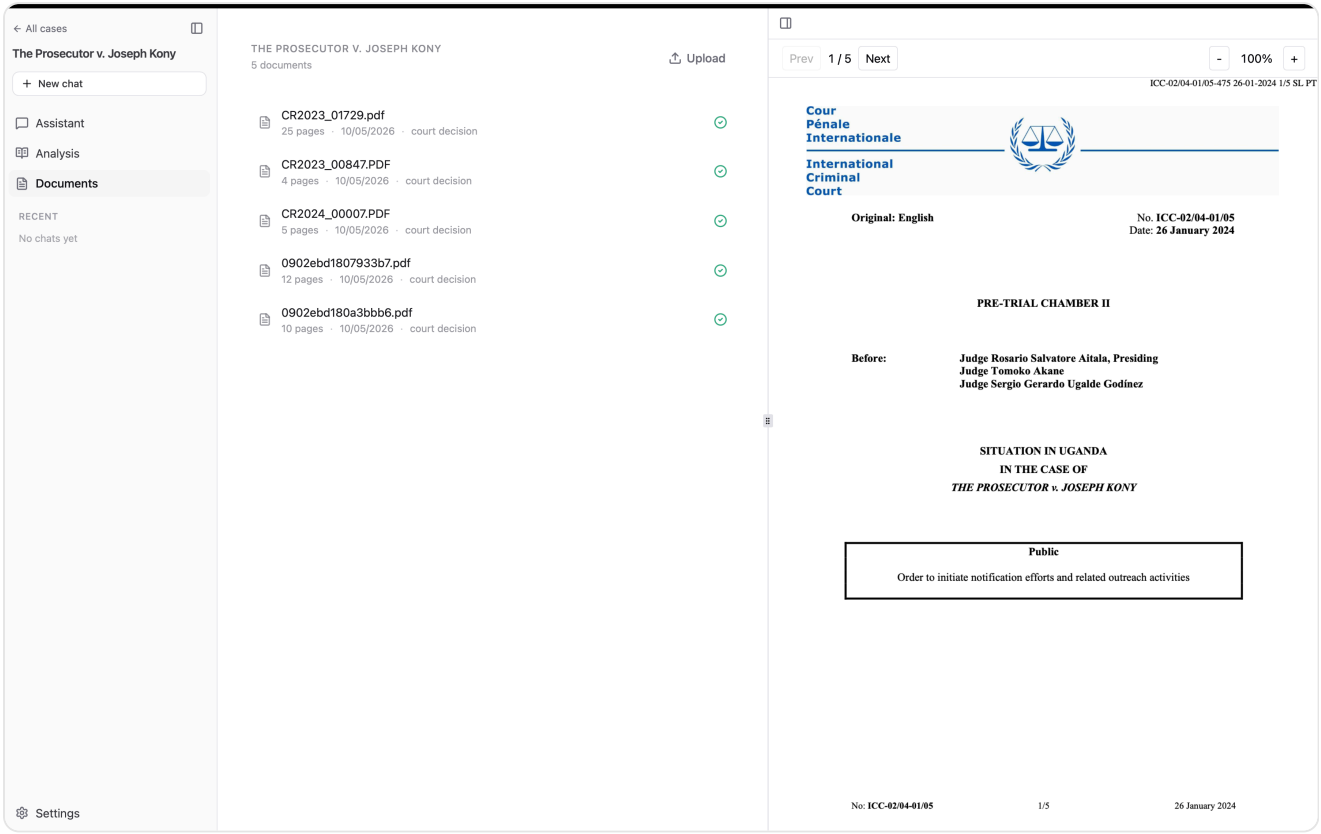
Challenge

The practice spent more than 60 hours analyzing every large case file: reading thousands of pages and tracking how names, dates, and evidence connected across them. Notes lived in post-its and Word docs scattered across the team, and cross-referencing 3,000-page files with Ctrl+F was no longer working. Generic AI tools could not handle the scale, structure, and detail sensitivity of a case file — ChatGPT could not cross-reference and cite thousands of pages of witness statements, court decisions, and scanned reports in a way the team could trust for legal work.

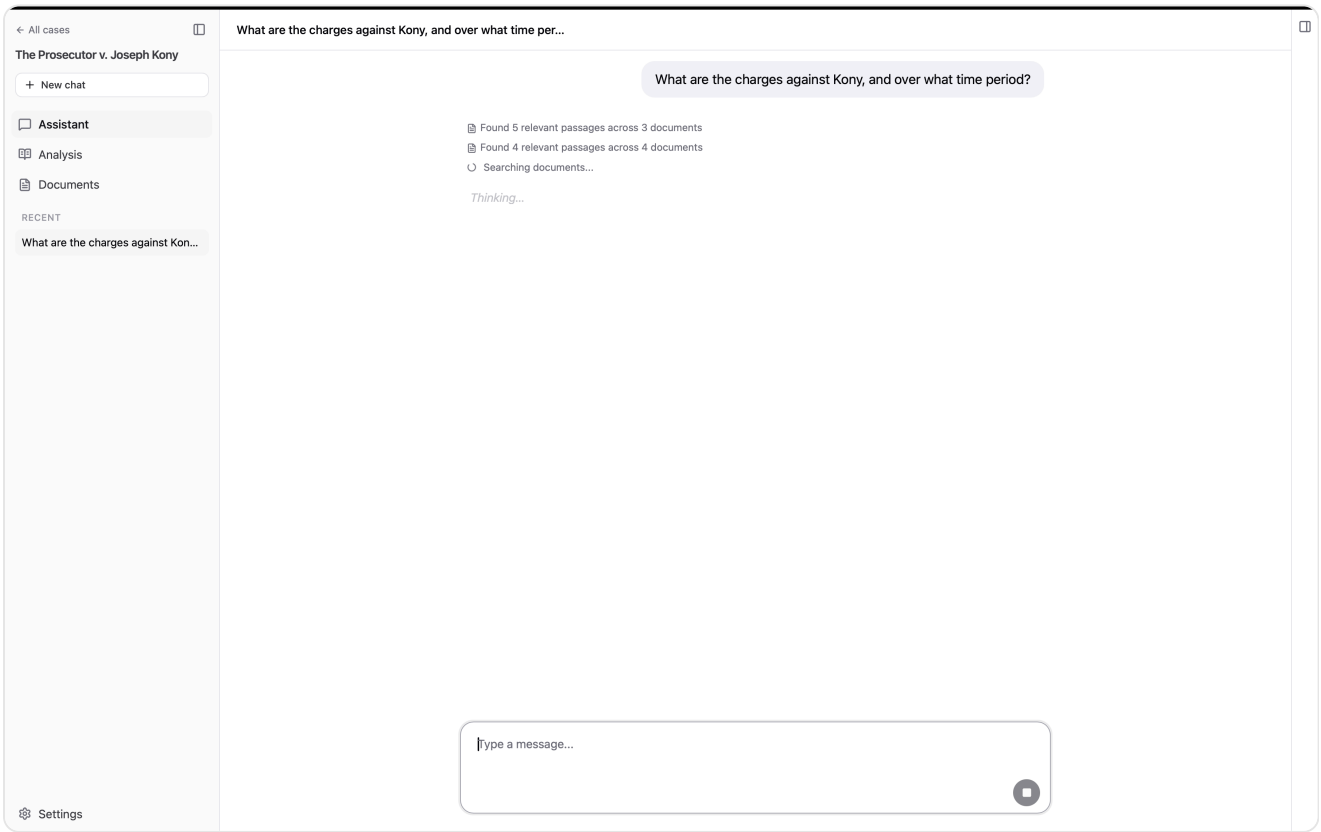
Approach

I built Casemap — case-analysis software shaped around the way the team reviews large criminal case files. The system ingests the full file — PDFs, scanned documents, handwritten notes, photographs — using computer vision and multimodal AI where needed, classifies each document by type, and indexes everything. The junior associate starts from AI-generated summaries and a working map of parties, roles, relationships, dates, events and communications, then asks questions across the file and receives answers with citations that open the original PDF in the side panel. After review, the system drafts a structured working case summary from the associate's highlights and annotations. **Legal judgment stays with the lawyers.**

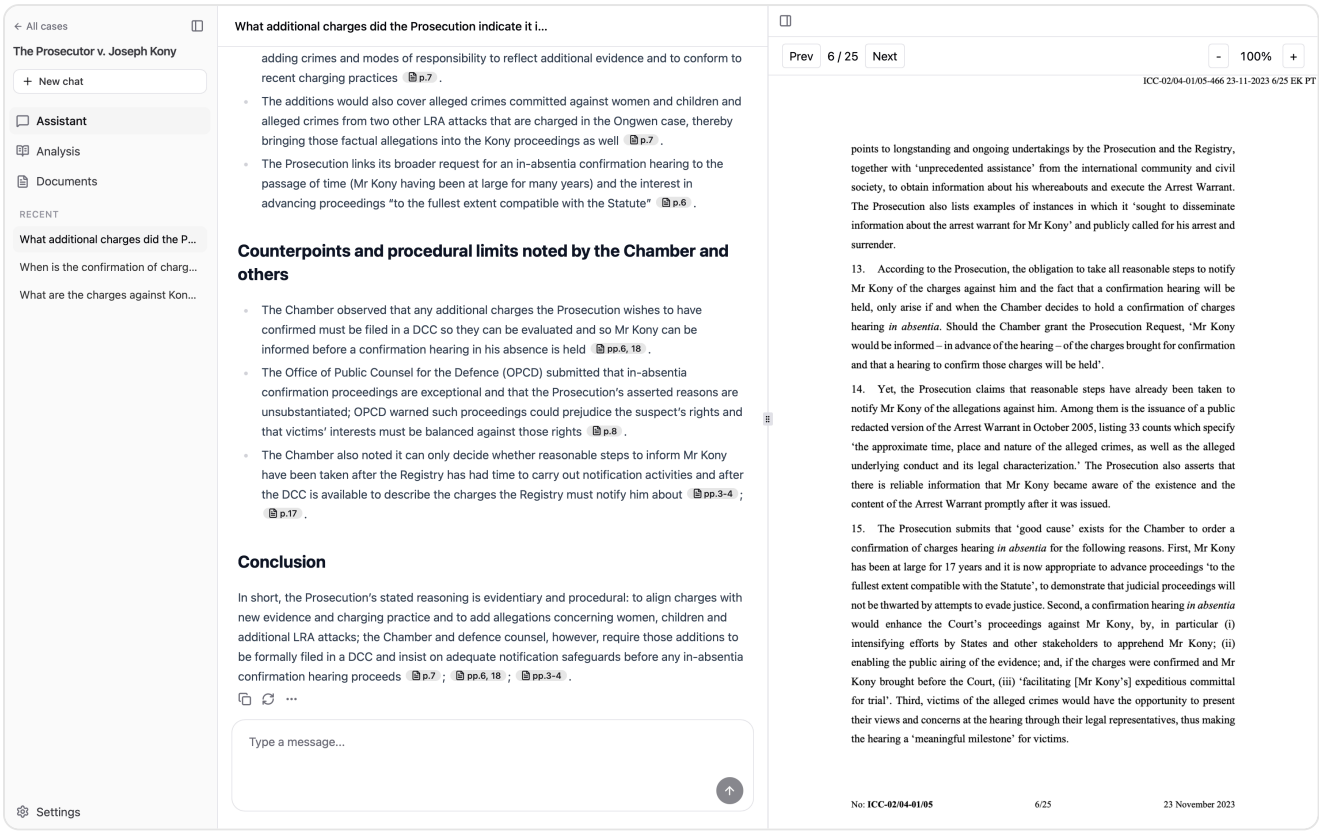
Solution Overview



All case documents in one place — classified, indexed and searchable from the start.



Ask questions across the full case file. The system searches the way a junior associate works through a case.



Every answer is cited. Click any citation to open the original PDF in the side panel.

How It Works

`Multimodal ingestion` `GraphRAG` `OCR` `Citation grounding` `Multi-agent retrieval`

Tech Stack

`FastAPI` `LangGraph` `Next.js` `PostgreSQL` `Azure`

Results & Impact

Workflow analysis and pilot observations put the saving at roughly 28 hours of case-preparation time per large case, without adding headcount or reducing human review. Long manual passes through the file became a guided review with cited answers and source verification. At the practice's current case volume, that would roughly double their large-case capacity per year.

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Saved per large case

2x*
Large-case capacity per year

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Project Details

LOCATION
Athens, Greece

TIMELINE
4 months (Q2–Q3 2026)

AGENT PLATFORM
Microsoft Foundry

TEAM SIZE
Solo

ENGAGEMENT MODE
Direct with the lawyers

ABOUT MANOS

Manos Fragkiadakis is an Applied AI Engineer building document intelligence and AI knowledge systems. He has over five years of Python software development experience and has built custom ERP and CRM systems for professionals. His recent work includes cited RAG across 3,000+ page legal case files and court-monitoring automation software used by 10+ law offices. He previously worked as an AI engineer and taught Natural Language Processing at Utrecht University.

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